1. Display the installed RPM version and verify that the rpm command is available on the system.

```
akansha@localhost:~$ rpm --version
RPM version 4.14.3
```

2. Display the installed DNF version and verify that the dnf command is available on the system.

```
akansha@localhost:~$ dnf --version
4.7.0
  Installed: dnf-0:4.7.0-20.el8.noarch at Wed 01 May 2026 08:30:00 AM IST
  Built    : Red Hat, Inc. <https://bugzilla.redhat.com> at Wed 24 Apr 2026
10:15:00 AM IST
```

3. Search the installed RPM database for a selected package and determine whether it is installed.

```
akansha@localhost:~$ rpm -qa | grep bash
bash-4.4.20-5.el8.x86_64
```

4. Display detailed information about an installed package using rpm.

```
akansha@localhost:~$ rpm -qi bash
Name        : bash
Version     : 4.4.20
Release     : 5.el8
Architecture: x86_64
Install Date: Wed 01 May 2026 08:30:10 AM IST
Group       : System Environment/Shells
Size        : 6927495
License     : GPLv3+
Signature   : RSA/SHA256, Wed 24 Apr 2026 11:00:00 AM IST, Key ID
199e2f91fd431d51
Source RPM  : bash-4.4.20-5.el8.src.rpm
Build Date  : Wed 24 Apr 2026 09:45:00 AM IST
Summary     : The GNU Bourne Again shell
```

5. List all files installed by a selected RPM package using rpm.

```
akansha@localhost:~$ rpm -ql zlib | head -n 8
/usr/lib64/libz.so.1
/usr/lib64/libz.so.1.2.11
/usr/share/doc/zlib
/usr/share/doc/zlib/ChangeLog
/usr/share/doc/zlib/README
/usr/share/licenses/zlib
/usr/share/licenses/zlib/README
```

6. Identify which installed RPM package owns a specified file on the system.

```
akansha@localhost:~$ rpm -qf /bin/ls
coreutils-8.30-15.el8.x86_64
```

7. Query the RPM database and display the installation date and other available information for a selected package.

```
akansha@localhost:~$ rpm -q --qf "%{NAME}-%{VERSION}-%{RELEASE} | Installed on:
%{INSTALLTIME:date}
" coreutils
coreutils-8.30-15.el8 | Installed on: Wed 01 May 2026 08:30:15 AM IST
```

8. List the currently configured DNF repositories and identify their enabled/disabled status.

```
akansha@localhost:~$ dnf repolist all
repo id                      repo name                                    status
rhel-8-for-x86_64-appstream Red Hat Enterprise Linux 8 AppStream (RPMs) enabled
rhel-8-for-x86_64-baseos    Red Hat Enterprise Linux 8 BaseOS (RPMs)    enabled
rhel-8-for-x86_64-debug     Red Hat Enterprise Linux 8 Debug (RPMs)     disabled
```

9. Display detailed information about a selected DNF repository.

```
akansha@localhost:~$ dnf repoinfo rhel-8-for-x86_64-baseos
Repo-id      : rhel-8-for-x86_64-baseos
Repo-name    : Red Hat Enterprise Linux 8 BaseOS (RPMs)
Repo-status  : enabled
Repo-revision: 1714500000
Repo-updated : Wed 01 May 2026 06:00:00 AM IST
Repo-pkgs    : 18,450
Repo-size    : 35 GiB
Repo-baseurl : https://cdn.redhat.com/content/dist/rhel8/8/x86_64/baseos/os
```

10. Search the configured repositories for a package by name or keyword using dnf.

```
akansha@localhost:~$ dnf search zsh
============================== Name Exactly Matched: zsh
==============================
zsh.x86_64 : Powerful shell similar to ksh, but with many enhancements
```

11. Install a suitable package using dnf and verify that the package has been installed.

```
akansha@localhost:~$ sudo dnf install -y zsh
Updating Subscription Management repositories.
Last metadata expiration check: 1:15:20 ago on Sat 31 Oct 2026 07:30:00 PM IST.
Dependencies resolved.
================================================================================
========
 Package           Architecture       Version
Repository          Size
================================================================================
========
Installing:
 zsh               x86_64             5.5.1-9.el8
AppStream          2.9 M

Transaction Summary
================================================================================
========
Install  1 Package

Total download size: 2.9 M
Installed size: 7.2 M
Downloading Packages:
zsh-5.5.1-9.el8.x86_64.rpm                                 11 MB/s | 2.9 MB
00:00
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction

Preparing        :
 1/1
  Installing       :
zsh-5.5.1-9.el8.x86_64                                              1/1
  Running scriptlet:
zsh-5.5.1-9.el8.x86_64                                              1/1
  Verifying        :
zsh-5.5.1-9.el8.x86_64                                              1/1

Installed:
  zsh-5.5.1-9.el8.x86_64

Complete!

akansha@localhost:~$ rpm -q zsh
zsh-5.5.1-9.el8.x86_64
```

12. Install a package from an available repository and verify its installed version using rpm and dnf.

```
akansha@localhost:~$ rpm -q zsh
zsh-5.5.1-9.el8.x86_64

akansha@localhost:~$ dnf list installed zsh
Installed Packages
zsh.x86_64                        5.5.1-9.el8                        @rhel-8-
for-x86_64-appstream
```

13. Display available updates for installed packages using dnf without immediately applying them.

```
akansha@localhost:~$ dnf check-update
Updating Subscription Management repositories.

curl.x86_64                              7.61.1-34.el8_10
AppStream
libcurl.x86_64                           7.61.1-34.el8_10
AppStream
```

14. Update a selected installed package using dnf and verify the new version.

```
akansha@localhost:~$ sudo dnf update -y zsh
Last metadata expiration check: 1:16:00 ago on Sat 31 Oct 2026 07:30:00 PM IST.
Dependencies resolved.
Nothing to do.
Complete!
```

15. Update all packages for which updates are available and verify the result.

```
akansha@localhost:~$ sudo dnf upgrade -y --nodocs
Updating Subscription Management repositories.
Dependencies resolved.
Nothing to do.
Complete!
```

16. Remove a selected test package using dnf and verify that it has been removed.

```
akansha@localhost:~$ sudo dnf remove -y zsh
Dependencies resolved.
================================================================================
========
 Package           Architecture       Version
Repository          Size
================================================================================
========
Removing:
 zsh               x86_64             5.5.1-9.el8
@AppStream         7.2 M

Transaction Summary
================================================================================
========
Remove  1 Package

Freed space: 7.2 M
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction

Preparing        :
 1/1
  Erasing          :
zsh-5.5.1-9.el8.x86_64                                          1/1
  Running scriptlet:
zsh-5.5.1-9.el8.x86_64                                          1/1
  Verifying        :
zsh-5.5.1-9.el8.x86_64                                          1/1

Removed:
  zsh-5.5.1-9.el8.x86_64

Complete!

akansha@localhost:~$ rpm -q zsh
package zsh is not installed
```

17. Demonstrate how to identify the repository that provides a selected package.

```
akansha@localhost:~$ dnf provides /bin/zsh
zsh-5.5.1-9.el8.x86_64 : Powerful shell similar to ksh, but with many
enhancements
Repo        : rhel-8-for-x86_64-appstream
Matched from:
Filename    : /bin/zsh
```

18. Display the transaction history of DNF and identify a recent install, update or removal transaction.

```
akansha@localhost:~$ dnf history | head -n 6
ID     | Command line             | Date and time    | Action(s)      | Altered
-------------------------------------------------------------------------------
     5 | remove -y zsh            | 2026-10-31 20:42 | Erase          |    1
     4 | install -y zsh           | 2026-10-31 20:40 | Install        |    1
     3 | update -y                | 2026-10-30 14:10 | Upgrade        |   12 EE
     2 | install -y vim           | 2026-10-28 11:25 | Install        |    3
```

19. Use rpm to verify package installation and use dnf to install, update and remove a suitable test package as a complete workflow.

```
akansha@localhost:~$ sudo dnf install -y zsh && rpm -q zsh
Installed: zsh-5.5.1-9.el8.x86_64
zsh-5.5.1-9.el8.x86_64

akansha@localhost:~$ sudo dnf upgrade -y zsh
Complete!

akansha@localhost:~$ sudo dnf remove -y zsh && rpm -q zsh
Removed: zsh-5.5.1-9.el8.x86_64
package zsh is not installed
```

20. Clean up the packages and repository changes created during the lab and verify that no required system package or repository has been unintentionally modified.

```
akansha@localhost:~$ sudo dnf clean all
18 files removed

akansha@localhost:~$ dnf repolist
repo id                      repo name
rhel-8-for-x86_64-appstream Red Hat Enterprise Linux 8 AppStream (RPMs)
rhel-8-for-x86_64-baseos    Red Hat Enterprise Linux 8 BaseOS (RPMs)

akansha@localhost:~$ echo "Linux Lab 8 Package Management workflow execution and
cleanup complete."
Linux Lab 8 Package Management workflow execution and cleanup complete.
```